

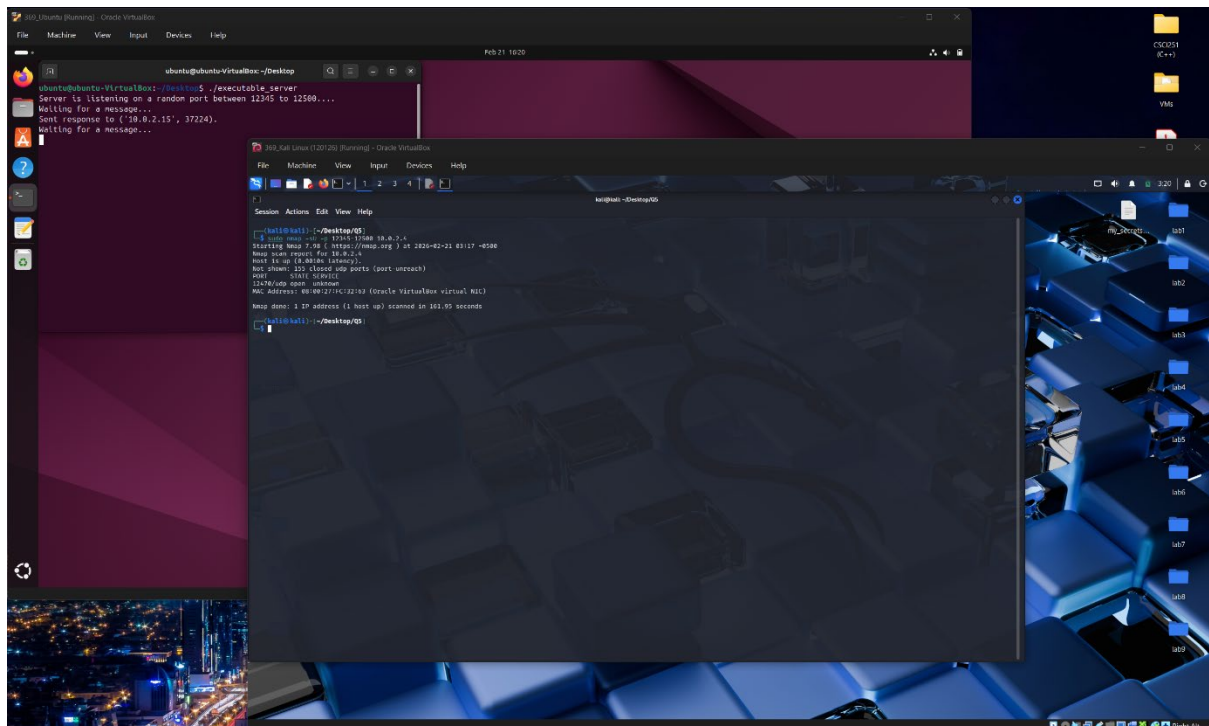
a) Identifying open port between 12345 and 12500

The server provides the voucher service using UDP on a port between 12345 and 12500. Therefore, 'nmap' tool was utilised, specifying the '-sU' flag to initiate a UDP scan as the standard TCP scans do not detect UDP services.

To optimise the scan speed, the scan was restricted to the known target range using the '-p' flag.

Command executed: 'sudo nmap -sU -p 12345-12500 10.0.2.4'

Screenshot of successful run:



b) Gift voucher code (MD5 Hash)

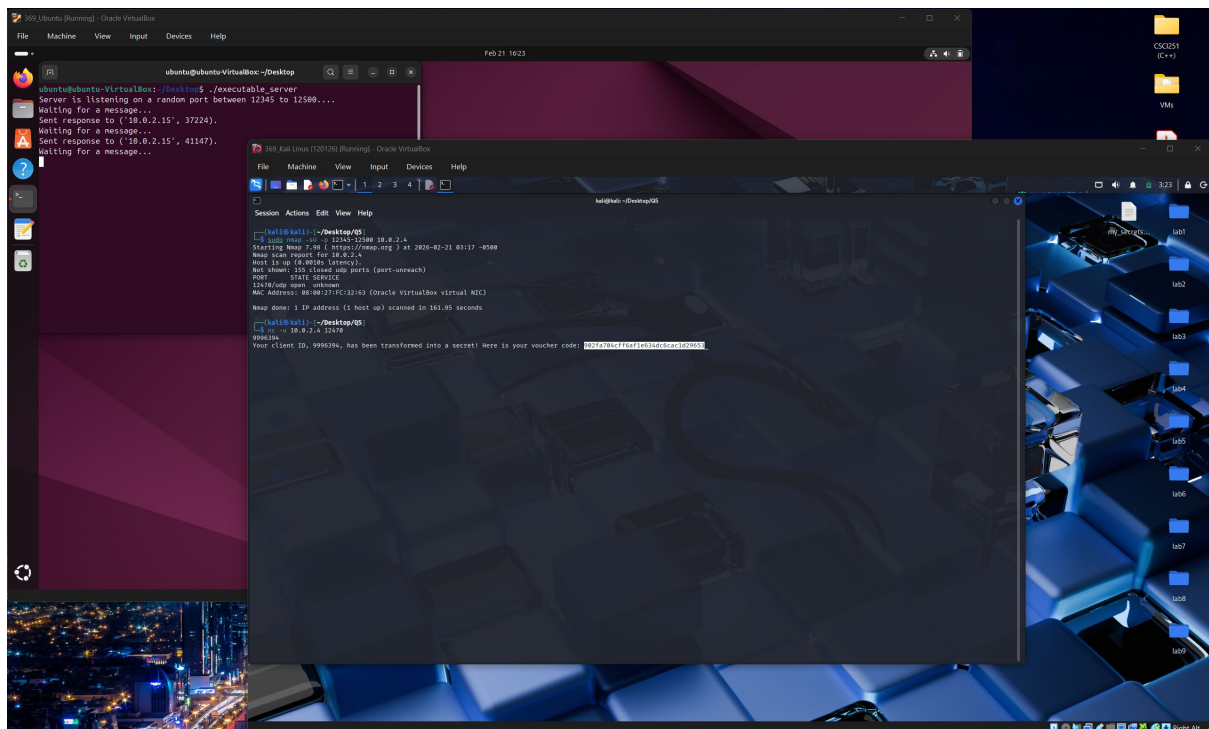
Client ID used: 9996394

Voucher code obtained (MD5 hash): 902fa704cff6af1e634dc6cac1d29653

To interact with the identified UDP service, netcat command, 'nc' was utilised. Because the service runs on UDP rather than the default TCP, the usage of the '-u' flag is mandatory to the netcat command so that it can establish a successful connection to the server. Upon successful connection, the 7-digit student ID as the client ID mentioned above was inputted into the terminal to receive the generated gift voucher code.

Command executed: 'nc -u 10.0.2.4 12470'

Screenshot of successful run:



c) Using Crunch and Hashcat

Cracked voucher plaintext: `tb9996394+$`

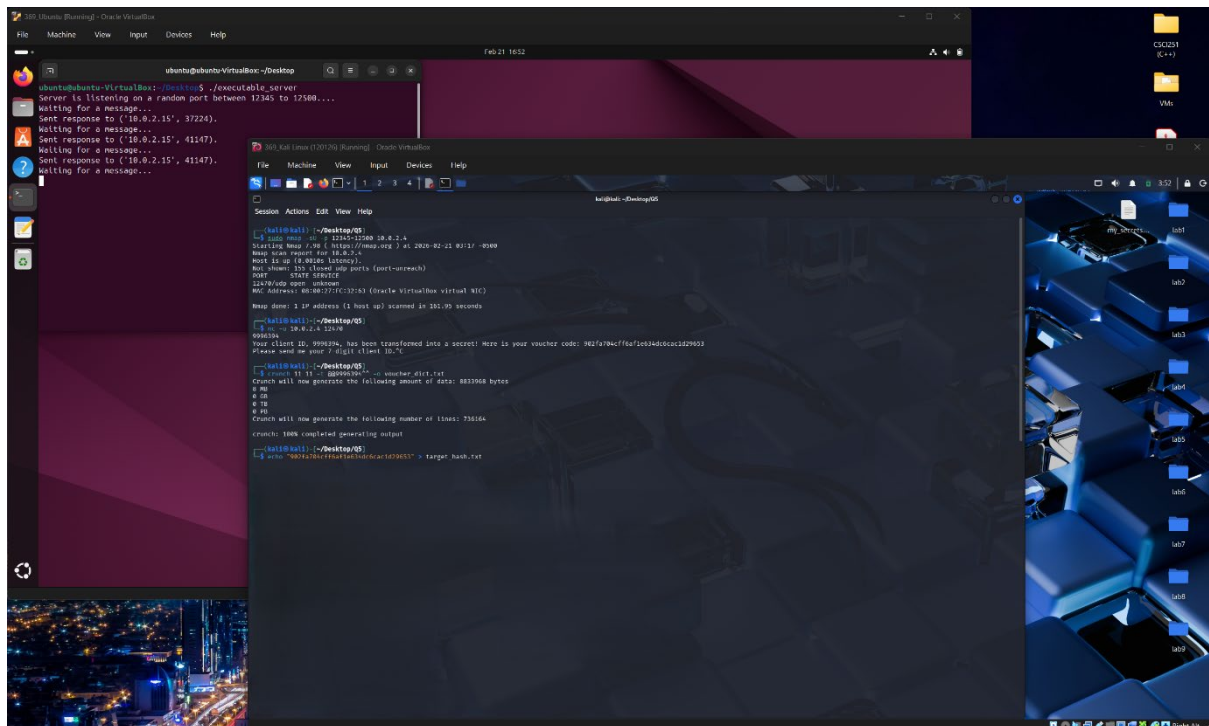
Identified Characters: A (2 lowercase letters): `'tb'`, B (2 symbols): `'+$'`

The server generates the voucher code using the MD5 hash function with the structure, ***VoucherCode = MD5(A||ClientID||B)***.

To crack the hash, a specific dictionary file, `'voucher_dict.txt'` containing every possible combination of the structure is required. The `'crunch'` tool was utilised to generate an 11-character wordlist. In crunch syntax, `'@'` represents lowercase letters and `'^'` represents symbols.

Command executed: `'crunch 11 11 -t @@9996394^^ -o voucher_dict.txt'`

Screenshot of successful run:



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Command executed: 'hashcat -m 0 target_hash.txt voucher_dict.txt'
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Screenshot of successful run:

